ASSET STORE USER DOCUMENTATION

Easy Line

Spline Mesh Deformation

and Prefab Placement for Unity

Offline documentation | Version 1.0

Render pipelines: Built-in, URP, HDRP

00 Product Overview

Easy Line is a Unity Editor and runtime toolkit for authoring cubic Bezier splines, deforming repeated geometry, placing complete prefabs, and moving objects along a path. It is designed for roads, rails, pipes, cables, barriers, tracks, tunnels, vines, and conveyor systems.

> Designed for production
> Work non-destructively while authoring, then keep the result procedural or bake it to a standard Unity prefab and mesh.

Compatibility

Item	Support
Unity version	See the Easy Line Asset Store listing for the currently supported Unity editor versions.
Render pipelines	Built-in, Universal Render Pipeline (URP), and High Definition Render Pipeline (HDRP).
Platforms	Generated geometry uses standard Unity meshes. Validate shaders, physics, and performance on your target platform.
Dependencies	No required third-party dependency. ProBuilder integration is optional and activates when ProBuilder is installed.
Documentation	This PDF is included locally in Assets/EasyLine and works offline.

Package Contents

Folder / File	Contents
Scripts	Runtime spline, deformation, placement, and follower components.
Editor	Custom Inspectors, Scene handles, draw tool, baking, and OBJ export.
Examples	Demo content for Built-in, URP, and HDRP.
Icons	Easy Line Scene tool icon.
EasyLine_Documentation.pdf	Complete offline user guide.

> Before importing examples
> Use the demo scene and materials that match your active render pipeline. The core tool itself generates standard Unity geometry.

01 Installation and Quick Start

Import from Unity Asset Store

1. Open Window > Package Manager in Unity.
2. Select My Assets in the Package Manager navigation panel.
3. Find Easy Line, then click Download if required.
4. Click Import. Keep the EasyLine folder structure intact.
5. Wait for Unity to import assets and compile scripts before opening a demo scene.

> Updating an existing project
> Back up or commit your project before importing an update. If you modified files inside Assets/EasyLine, move those changes outside the package folder first.

Create Your First Road

1. Create an empty GameObject and add BezierSpline.
2. Select it and activate the EasyLine Draw Tool in the Scene view toolbar.
3. Click a surface to place anchors. Click and drag to shape the tangent while placing a point.
4. Add SplineMeshDeformer and assign the spline to Spline Path.
5. Choose Mesh or Prefab source mode, assign your source asset, and select its local Forward Axis.
6. Assign materials, set Segment Count, and enable Auto Live. Use Stretch to Fit Curve or a small negative Overlap Offset to close gaps.

Source Asset Checklist

- Forward Axis: identify the source model's local length axis.
- Pivot: use a predictable pivot aligned with the modular section.
- Read/Write: enable mesh readability when runtime deformation requires CPU mesh access.
- Materials: use shaders that match the active render pipeline.

02 Drawing and Editing Splines

EasyLine Draw Tool

Input	Action
Left-click on a surface	Adds an anchor or completes the first segment.
Left-click and drag	Adds a point and defines its curve tangent.
Left-click a point	Drags an anchor or control handle.
Click near a segment	Inserts a new point into the existing curve.
Click the first anchor	Closes a spline with at least two segments.
Enter	Finishes drawing and restores the previous tool.
Escape	Finishes the active spline; a later Escape restores the previous tool.

Starting another path
Clicking far from the active spline end can create a new spline object. The current implementation uses a threshold of 100 world units.

BezierSpline Inspector

Control	Purpose
Loop Curve	Connects the last anchor to the first and synchronizes the loop seam.
+ Add Segment	Appends another cubic Bezier segment.
- Remove Last	Removes the final segment while retaining at least one.
Smooth All Tangents	Aligns handles for a smoother path.
Reset All Scales	Restores every anchor scale to 1,1,1.
Reset Curve	Restores the default four-point straight spline.

Anchors, Scale, and Banking

White points are anchors; the intermediate handles define the segment shape. Anchor scale is interpolated along the path, allowing roads and pipes to widen or taper. Anchor roll controls banking around the path direction.

- Scale Tool (R): changes the selected anchor scale.
- Rotate Tool (E): edits the selected anchor's roll/banking.
- On a loop, the first and last anchors represent the same seam. Check tangent, scale, and roll continuity.

Authoring tip
Use as few anchors as the shape allows. Establish the route first, refine tangents second, and add scale and banking last.

03 Spline Mesh Deformer

SplineMeshDeformer builds one output mesh from a repeated source or a set of indexed layers. It can bend or rigidly place geometry, generate collision, animate the result, and save a prefab or OBJ.

Geometry Source

Field	Description
Spline Path	BezierSpline that defines the output path.
Source Mode: Mesh	Direct Mesh source; assign Materials manually.
Source Mode: Prefab	Reads meshes and materials from a prefab hierarchy.
Use Mixed Meshes - LAYERS	Enables separate sources for selected segment ranges.
Longitudinal Cut	Adds divisions along the spline axis before deformation; range 0-4.

> ProBuilder
> When ProBuilder is installed, Easy Line can rebuild geometry directly from ProBuilderMesh. A ProBuilder prefab may have missing geometry if the package is unavailable.

Array Settings

Field	Use
Segment Count	Number of source copies. With Stretch to Fit, it becomes a density control.
Forward Axis	Z, X, Y, or a negative variant; choose the source asset's length axis.
Overlap Offset	Spacing correction. Negative values close small seams.
Mesh Scale	Scale applied to every source copy.
Deform Mesh	Bends vertices; disabled means rigid placement along the path.

Choosing Forward Axis

If the result runs sideways, appears flattened, or stretches across the path, try Z, X, and Y. If the axis is correct but direction is reversed, choose NegativeZ, NegativeX, or NegativeY. Do not compensate for the wrong axis with extreme scaling.

Longitudinal Cut

Subdivision adds geometry before bending, allowing long low-density modules to follow curves smoothly. Use the lowest level that removes visible faceting. Mixed Mesh mode provides a separate setting for every layer.

03 Spline Mesh Deformer - Advanced

Option	Effect
Merge Vertices	Welds matching boundary vertices between repeated modules.
Merge Distance	Welding threshold. Excessive values may damage sharp detail.
Smooth Normals	Blends normals across segment boundaries and the loop seam.
Curve Bend Density	0 gives even spacing; positive values shrink around bends, negative values stretch.
Smooth Bend Strength	Softens rapid direction changes.
Stretch to Fit Curve	Fits road segments to the complete curve without a gap or overshoot.
Generate World UVs	Rebuilds UVs from world position for continuous textures.
World UV Scale	Scale multiplier for world-space UV projection.

Recipe: A Seamless Road

1. Enable Deform Mesh and Stretch to Fit Curve.
2. Enable Merge Vertices and Smooth Normals.
3. Begin with a small Merge Distance; the default is 0.001.
4. Increase Longitudinal Cut by one level if curves still look faceted.
5. If texture seams remain, use tiled materials or Generate World UVs.

Live and Manual Updates

Auto Live updates the mesh after Inspector or spline changes. Disable it while editing a large track, then press Deform Now. This avoids repeated rebuilds while entering multiple values.

> Subdivision diagnostic
> The Inspector displays the output vertex count. It should increase when Longitudinal Cut is raised. If it does not, verify the active source and Mixed Mesh mode.

04 Mixed Mesh Layers

Mixed Mesh mode assigns different meshes or prefabs to segment ranges. It supports combinations such as road - bridge - road, pipe - valve - pipe, or a track with dedicated props and end pieces.

> **Priority rule**
> When ranges overlap, the element lower in the list overrides elements above it. List order is part of the configuration.

Range and Source

Field	Meaning
Start Index / End Index	Inclusive segment range. For example, 3-5 covers 3, 4, and 5.
Mode	Mesh or Prefab source for this layer.
Material Overrides	Optional replacement materials.
Deformation & Collider	Auto, Road, or Prop classification.
Subdivisions	Longitudinal subdivision level for this layer only.

Road, Prop, and Auto

Mode	Behavior
Road	Continuous geometry that can bend along the spline.
Prop	Rigid by default; suitable for posts, lights, signs, and similar objects.
Auto	Uses a length/shape heuristic to classify the element.

Per-Layer Constraints

Option	Use
Stretch To Index Ends	Scales along the forward axis to fill the assigned range exactly.
Simplify Collider To Box	Simplifies only the prop collider and requires the global collider options.
Local Scale / Flip Axis	Adjusts or mirrors geometry without editing the source asset.
Position / Rotation Offset	Corrects the layer's local placement.
Force 100% Upright	Ignores spline banking and stays vertical.
Deform Props	Allows a Prop-classified object to bend.
Lock Rotation X/Y/Z	Locks pitch, yaw, or roll for the selected layer.

05 Spline Prefab Instantiator

The instantiator places complete GameObjects, preserving scripts, lights, VFX, and prefab hierarchy. Its hybrid mode can bend the child meshes of every generated instance.

Core Settings

Field	Description
Spline Path	Placement curve.
Source Prefab	Prefab to instantiate.
Spacing Mode	Total Count or Fixed Distance.
Instance Count	Number of objects in Total Count mode.
Spacing Distance	World-space interval in Fixed Distance mode.
Global Offset	Shifts the full distribution along the spline.
Position / Rotation Offset	Per-instance transform correction.
Base Scale	Shared starting scale.
Forward Axis	Prefab axis that points along the path.

Alignment and Hybrid Deformation

Option	Meaning
Align to Path	Rotates instances to match the spline direction.
Stay Upright (No Roll)	Keeps instances vertical and ignores banking.
Deform Meshes (Hybrid)	Bends child meshes while preserving other prefab components.
Smart Stretch to Fit	Scales mesh length to the spacing between instances.
Apply Spline Scale	Applies per-anchor scale; disabling it preserves original prefab shape.

> Inspector note
> Apply Spline Scale exists in the component and is used by the implementation, but the current custom Inspector does not draw it. Set it through Debug Inspector or code when required.

Variation

- Position Jitter: random position offset on each axis.
- Random Rotation: random Euler rotation range.
- Scale Randomness: 0.5 means approximately +/-50% size variation.

Auto Live rebuilds instances after edits. For heavy prefabs, disable it, configure the values, and press Spawn Now. The Clear button removes generated objects.

06 Spline Follower

SplineFollower moves a Transform along a path in Play Mode. Current Distance can also preview its position in the Editor. Distance mapping provides consistent world-space speed across uneven Bezier segments.

Field	Description
Spline	Path to follow.
Speed	World units per second; a negative value reverses travel.
Follow Mode: Loop	Wraps distance at the end.
Follow Mode: Once	Clamps distance at the start or end.
Rotation: None	Leaves rotation unchanged.
Rotation: LookAhead	Faces the curve tangent.
Rotation: SplineRotation	Builds a stable orientation from tangent and an up vector.
Use Constant Speed	Maps physical distance to parameter t.
Current Distance	Current world-space distance along the path.

Code Example

```
using EasyLine; using UnityEngine; public class StartVehicle : MonoBehaviour { [SerializeField] private SplineFollower follower; public void Begin(float speed) { follower.speed = speed; follower.currentDistance = 0f; follower.RefreshMapper(); } }
```

Runtime curve changes
Call RefreshMapper() after changing spline points at runtime so length and constant-speed mapping are recalculated.

Rotation Mode Guidance

- Start with LookAhead for vehicles and cameras.
- Use None when another system owns rotation.
- SplineRotation creates a stable orientation from the tangent and world up; it does not directly apply anchor banking.

07 Animation, Physics, and Performance

Conveyor Animation

Option	Meaning
Animate Conveyor	Scrolls output geometry along the path in Play Mode.
Conveyor Speed	Animation speed and direction.
Fast Animation Mode	Skips expensive smoothing work on every frame.
Static Collider	Prevents MeshCollider recooking every animation frame.

Recommended default
Keep Fast Animation Mode and Static Collider enabled for conveyor belts unless gameplay requires a physically moving collision surface.

Mesh Collider

Option	Use
Generate Mesh Collider	Creates or updates collision from Easy Line geometry.
Simplification 1-20	1 is full detail; higher values skip more collider segments.
Simplify Props as Boxes	Replaces permitted prop collision with simple boxes.

Performance Checklist

Symptom	First Optimization
Slow Inspector	Disable Auto Live; reduce Segment Count; edit in batches.
Large output mesh	Reduce Longitudinal Cut and unnecessary layers.
Play Mode CPU spikes	Enable Fast Animation Mode and Static Collider.
Expensive physics	Increase collider simplification; use boxes for props.
Heavy prefab placement	Disable hybrid deformation or reduce instance count.
Lighting artifacts	Try Smooth Normals before increasing subdivision.

Subdivision Cost

Each level adds more geometry along the spline axis. Cost scales with segment and layer count, so levels 3-4 should be reserved for close-up assets or sources that genuinely lack enough vertices to bend smoothly.

08 Export and Baking

Bake to Prefab

Creates a static Unity prefab and mesh from the current deformed result. Use it when the path is final and no longer needs procedural updates.

1. Finish spline editing and press Deform Now.
2. Verify materials, seams, collider, and orientation.
3. Press Bake to Prefab and choose a save location.
4. Place the saved prefab in a clean scene and verify it independently.

Export to OBJ

Exports the current geometry as OBJ for Blender, Maya, 3ds Max, or another DCC tool. OBJ is a static snapshot and does not contain Easy Line behavior.

> Before saving
> Export uses the current component state. If Auto Live is disabled, always press Deform Now immediately before Bake or Export.

Procedural or Static?

Keep Procedural	Bake Static
The spline changes at runtime.	The path is final and immutable.
Conveyor animation uses the component.	Minimum scene CPU cost is important.
Parametric editing must remain available.	The asset is ready for lightmaps, prefab variants, or DCC work.

Post-Bake Checks

- Are all submesh materials present?
- Is Transform scale predictable for physics and lighting?
- Does the baked object contain any unnecessary active deformer?
- Does it render correctly in the target pipeline?

09 Troubleshooting

Symptom	Cause and Fix
Mesh runs sideways	Wrong Forward Axis. Choose the local length axis or its Negative variant.
No geometry	Missing source, no MeshFilter in prefab, or unavailable ProBuilder data.
Gaps between modules	Enable Stretch to Fit, use a small negative Overlap Offset, and inspect the pivot.
Hard lighting seams	Enable Merge Vertices and Smooth Normals; raise Merge Distance carefully.
Faceted bends	The source lacks lengthwise vertices. Increase Longitudinal Cut.
Props bend like roads	Set the layer to Prop and disable Deform Props.
Lamp tilts on turns	Enable Force 100% Upright or Stay Upright.
Collider is too heavy	Increase simplification and use box colliders for props.
Mesh fails in Play Mode	Enable Read/Write or use Fix Mesh Readability Now.
Changes are not visible	Enable Auto Live or press Deform Now / Spawn Now.
Visible loop seam	Check first/last tangents, scale, and roll; enable Smooth Normals.
Variation keeps changing	The instantiator rebuilt its list. Disable Auto Live while tuning.

Diagnostic Order

1. Confirm Spline Path and the source asset are assigned.
2. Set the correct Forward Axis.
3. Temporarily disable Mixed Meshes, collider generation, and animation.
4. Re-enable features one at a time: deformation, stretch, smoothing, collider, animation.
5. For Play Mode-only failures, inspect mesh readability, collider updates, and Fast Animation Mode.

10 API Reference and Checklist

Key Methods

Type	Method	Purpose
BezierSpline	GetPoint(float t)	Returns a local spline point for t from 0 to 1.
BezierSpline	GetDirection(float t)	Returns the normalized tangent direction.
BezierSpline	GetRoll / GetScale	Returns interpolated banking or anchor scale.
BezierSpline	GetMapper(...)	Creates the physical-distance-to-t mapper.
BezierSpline	MarkDirty()	Invalidates cached mapping after point changes.
SplineMeshDeformer	Deform(bool)	Rebuilds the generated mesh.
SplineMeshDeformer	BakeToPrefab()	Saves a static prefab and mesh.
SplineMeshDeformer	ExportToOBJ()	Exports the current result as OBJ.
SplinePrefabInstantiator	UpdateInstances()	Rebuilds placed instances.
SplinePrefabInstantiator	ClearInstances()	Removes generated instances.
SplineFollower	RefreshMapper()	Recalculates distance mapping.

Minimal Spline Sampling

```
BezierSpline spline = GetComponent(); float t = 0.35f; Vector3 worldPoint =
spline.transform.TransformPoint(spline.GetPoint(t)); Vector3 worldDirection =
spline.transform.TransformDirection(spline.GetDirection(t));
```

Delivery Checklist

- Spline route, loop seam, banking, and scale are correct.
- Forward Axis is correct without compensating through extreme scale.
- Materials match the target render pipeline.
- Longitudinal Cut is as low as visual quality allows.
- Auto Live is disabled where a static result has already been baked.
- Collider detail is justified and is not rebuilt unnecessarily.
- The result was tested in Play Mode and in the target build.

End of documentation

This guide describes the Easy Line code included in this project. Update it whenever Inspector fields, deformation behavior, or export workflows change.

11 Support, Updates, and Known Limitations

This offline guide is the primary reference included with the Asset Store package. For the latest product information, use the online guide or contact the publisher directly.

Contact and Online Documentation

Email Support
ripvertices96@gmail.com

Easy Line Sites Guide
Open online documentation

Requesting Support

Include enough information to reproduce the issue:

- Unity editor version and operating system.
- Active render pipeline and pipeline package version.
- Easy Line version shown in your Asset Store package.
- A short list of reproduction steps and the expected result.
- Relevant Console errors, screenshots, and a minimal test scene when possible.

Updating Easy Line

1. Back up or commit the project.
2. Open Window > Package Manager > My Assets.
3. Select Easy Line, download the update, and import it into the project.
4. Verify spline geometry, materials, colliders, baking, and runtime animation.

Do not edit package files directly
Keep custom scripts, prefabs, and materials outside Assets/EasyLine so updates cannot overwrite project-specific work.

Known Technical Considerations

Area	Consideration
Runtime mesh deformation	Source meshes must allow CPU access. Use Fix Mesh Readability Now or enable Read/Write in import settings.
High-poly sources	Subdivision, high Segment Count, and hybrid prefab deformation increase memory and rebuild cost.
Animated colliders	Keep Static Collider enabled unless gameplay requires collision to change every frame.
Prefab spline scale	Apply Spline Scale can be changed through code or the Debug Inspector.
Pipeline materials	Demo materials are pipeline-specific even though generated geometry is pipeline agnostic.

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